Solution to Challenge 10

The Lakes of Wada

(Alligood, Sauer & Yorke, *Chaos: An Introduction to Dynamical Systems*, Ch. 10)

Overview. A *basin cell* Q is a trapping region whose boundary is made of pieces of the stable and unstable manifolds of a saddle periodic orbit $\mathbf{p}$. Steps 1–3 set up the basic topology of trapping regions and cells. Steps 4–6 prove that every boundary point of $\mathrm{bas}(Q)$ which can be reached by a curve from inside the basin lies on the stable manifold of $\mathbf{p}$ (Proposition 1). Step 7 shows such points are dense in the boundary (Proposition 2). Step 8 shows that if the unstable manifold of $\mathbf{p}$ pokes into another basin B_2, the whole boundary of $\mathrm{bas}(Q)$ belongs to the boundary of B_2 (Proposition 3), and hence to the boundaries of two other basins when it pokes into two of them: a Wada basin (Corollary 4).

Conventions and standing facts

Topology. For $A \subset \mathbb{R}^2$: $\operatorname{int} A$, $\operatorname{cl} A$, $\operatorname{bd} A$ are the interior, closure and boundary. If B is open then $\operatorname{bd} B = \operatorname{cl} B \setminus B$; if Q is closed then $\operatorname{bd} Q = Q \setminus \operatorname{int} Q$. We write $x_n = F^n(x)$. A *curve* with endpoint x is a continuous map $\gamma : [0,1] \to \mathbb{R}^2$ with $\gamma(0) = x$ (its image is also called the curve).

The map. As throughout the chapter, F is a *diffeomorphism* of $\mathbb{R}^2$ (smooth, one-to-one, onto, with smooth inverse; this is what stable-manifold theory needs). In particular F, F^{-1} are homeomorphisms, so $F(\operatorname{int} A) = \operatorname{int} F(A)$, $F(\operatorname{bd} A) = \operatorname{bd} F(A)$, $F(\operatorname{cl} A) = \operatorname{cl} F(A)$, and the same for F^{-1}.

Saddle orbit. $\mathbf{p} = \{p_1, \dots, p_k\}$ is a saddle periodic orbit, $F(p_i) = p_{i+1}$ (indices mod k). Put $K = 2k$, so that F^K fixes every p_i and maps each of the two branches of $W^s(p_i)$ and of $W^u(p_i)$ to itself. Write $W^s(\mathbf{p}) = \bigcup_i W^s(p_i)$, $W^u(\mathbf{p}) = \bigcup_i W^u(p_i)$. Standard facts (Stable Manifold Theorem):

(M1) $W^s(\mathbf{p})$ and $W^u(\mathbf{p})$ are F-invariant; $x \in W^s(\mathbf{p})$ iff $x_n \to \mathbf{p}$; consequently $F^n(x) \in W^s(\mathbf{p})$ for some $n \geq 0$ implies $x \in W^s(\mathbf{p})$. Each branch is an injectively immersed half-line $\beta \cong [0, \infty)$ with the point p_i at parameter 0.

(M2) *Dynamics on a branch.* Let β be a branch of $W^u(p_i)$ with parameter $s \in [0, \infty)$ and $g = F^{-K}|_\beta$. Then g is a continuous injective map of $[0, \infty)$ into itself with $g(0) = 0$, hence increasing, and $g^n(s) \to 0$ for all s. So $g(s) < s$ for $s > 0$ (if $g(s_0) \geq s_0$, monotonicity gives $g^n(s_0) \geq s_0$ for all n, contradicting $g^n(s_0) \to 0$). Thus

$$F^{-K}\big([0,e]\big) \subset [0,e], \qquad F^K(s) > s, \qquad F^{Kn}(s) \to \infty \quad (s > 0). \tag{1}$$

(M3) *Lambda Lemma.* Let g be a diffeomorphism with a saddle fixed point q, and let D be an arc crossing $W^s_g(q)$ transversally at a point. Then for every $y \in W^u_g(q)$ and $\delta > 0$ there is n with $g^n(D) \cap N_\delta(y) \neq \emptyset$.

Cells. A cell Q generated by $\mathbf{p}$ is a closed set homeomorphic to a closed disk (so $\operatorname{int} Q \neq \emptyset$, Q is bounded, and $\operatorname{int} Q$, being homeomorphic to an open disk, is path connected) with $\operatorname{bd} Q = Q_s \cup Q_u$ (disjoint) as in the text; I use the following features of the definition (see the figure on p. 435: the stable sides pass through the points p_i, the unstable sides come from loops of $W^u(p_i)$):

(C1) $Q_s \subset W^s(\mathbf{p})$ consists of the points p_i and the *stable sides* $\sigma_i \subset W^s(p_i)$, arcs with p_i in their relative interior;
(C2) $Q_u \subset W^u(\mathbf{p})$ is the union of the (open) *unstable sides*, one for each i, lying on a branch of $W^u(p_i)$;
(C3) $F(Q_s) \subset Q_s$: stable sides are carried toward $\mathbf{p}$ along the stable manifold (this is part of what "cell generated by $\mathbf{p}$" means; it is the only property of the stable sides used besides (C1)).

I also use, as the text allows, the fact that for a cell $F(\operatorname{bd} Q) \subset Q \Rightarrow F(Q) \subset Q$.

Basins. $\operatorname{bas}(Q) = \{x : \ x_n \in \operatorname{int} Q \text{ for some } n \geq 0\} = \bigcup_{n\geq 0} F^{-n}(\operatorname{int} Q)$. I write $B = \operatorname{bas}(Q)$.

Step 1: Trajectories that enter a trapping region stay in it

Let Q be a trapping region: $F(Q) \subset Q$. Suppose $x_j \in Q$. Since $A \subset B'$ implies $F(A) \subset F(B')$,

$$x_{j+1} = F(x_j) \in F(Q) \subset Q,$$

and by induction $x_n \in Q$ for all $n \geq j$. ■

Interior version (used later). The same is true for $\operatorname{int} Q$: $F(\operatorname{int} Q) = \operatorname{int} F(Q)$ is open and contained in $F(Q) \subset Q$, and $\operatorname{int} Q$ is the largest open subset of Q, so $F(\operatorname{int} Q) \subset \operatorname{int} Q$. Hence once $x_j \in \operatorname{int} Q$, $x_n \in \operatorname{int} Q$ for all $n \geq j$. (This is the statement "those trajectories then remain in $\operatorname{int} Q$" in the text.)

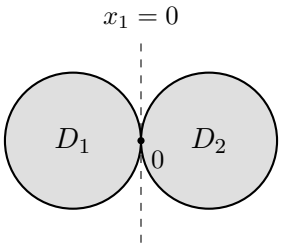
Step 2: Topological properties of basins

(1) $\operatorname{bas}(Q)$ is open

Let $x \in B$ and choose $j \geq 0$ with $F^j(x) \in \operatorname{int} Q$. Since F^j is continuous and $\operatorname{int} Q$ is open, $V = (F^j)^{-1}(\operatorname{int} Q)$ is an open set containing x. Every $y \in V$ has $F^j(y) \in \operatorname{int} Q$, so $y \in B$. Thus $V \subset B$ and B is open. ■

(2) $\operatorname{bas}(Q)$ is path connected

The statement needs $\operatorname{int} Q$ path connected. If only Q is assumed path connected the claim can fail.



Counterexample. Let $F(x) = x/2$ and $Q = D_1 \cup D_2$, two closed unit disks centered at $(\mp 1, 0)$, tangent at the origin. Then Q is bounded, Q is path connected, $\operatorname{int} Q \neq \emptyset$, $F(Q) \subset Q$ (each D_i is convex and contains 0), and $F(Q) \neq Q$: a trapping region. But $x_n = x/2^n$ lies in $\operatorname{int} D_1$ for large n iff $x_1 < 0$ (the condition $(x_1/2^n + 1)^2 + (x_2/2^n)^2 < 1$ reads $2x_1 + (x_1^2 + x_2^2)/2^n < 0$), similarly in $\operatorname{int} D_2$ iff $x_1 > 0$, and never in $\operatorname{int} Q$ if $x_1 = 0$. So $\operatorname{bas}(Q) = \{x_1 < 0\} \cup \{x_1 > 0\}$ is not connected.

Corrected claim. If $\operatorname{int} Q$ is path connected (true for every cell, being an open disk), then $\operatorname{bas}(Q)$ is path connected.

Proof. Let $x, y \in B$ with $F^j(x) \in \operatorname{int} Q$ and $F^l(y) \in \operatorname{int} Q$. With $m = \max(j, l)$, Step 1 (interior version) gives $F^m(x), F^m(y) \in \operatorname{int} Q$. Choose a path $s : [0,1] \to \operatorname{int} Q$ from $F^m(x)$ to $F^m(y)$ and put $\sigma = F^{-m} \circ s$. Then σ is continuous (as F^{-m} is), $\sigma(0) = x$, $\sigma(1) = y$, and $F^m(\sigma(\tau)) = s(\tau) \in \operatorname{int} Q$, so $\sigma(\tau) \in B$ for every τ. So σ is a path in B from x to y. ■

(Another sufficient condition: $F(Q) \subset \operatorname{int} Q$ and Q path connected, since then $Q \subset B$ and the same pull-back works.)

Step 3: A cell is a trapping region iff $F(Q_u) \subset Q$

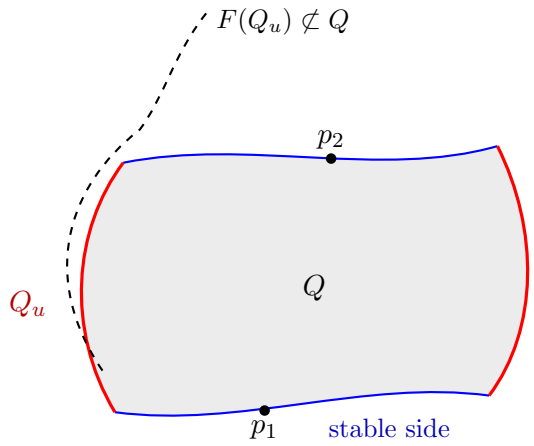
($\Rightarrow$) If Q is a trapping region then $Q_u \subset \operatorname{bd} Q \subset Q$ (Q is closed), so $F(Q_u) \subset F(Q) \subset Q$.

($\Leftarrow$) Assume $F(Q_u) \subset Q$. By (C3), $F(Q_s) \subset Q_s \subset Q$. Therefore

$$F(\operatorname{bd} Q) = F(Q_s) \cup F(Q_u) \subset Q,$$

and by the fact quoted in the text, $F(Q) \subset Q$. Since Q is bounded with nonempty interior, it remains to see $F(Q) \neq Q$. Suppose $F(Q) = Q$. Then $F(\operatorname{bd} Q) = \operatorname{bd} Q$. Let $\bar{u} = [s, e]$ be the closure of an unstable side, an arc of a branch of $W^u(p_j)$. Then $F(\bar{u})$ is a connected subset of $\operatorname{bd} Q \cap W^u(p_{j+1})$. The latter consists of the closed unstable side on $W^u(p_{j+1})$ plus isolated points where stable sides cross $W^u(p_{j+1})$ transversally, so $F(\bar{u})$ lies in the closed unstable side of p_{j+1}. Applying this K times, $F^K(\bar{u}) \subset \bar{u}$, i.e. $[F^K(s), F^K(e)] \subset [s, e]$ in the parameter of the branch, which contradicts $F^K(e) > e$ in (1). So $F(Q) \neq Q$ and Q is a trapping region. ■

A cell that is not a trapping region. The condition $F(Q_u) \subset Q$ is a genuine restriction: F stretches unstable arcs ((1)), so if the branch of W^u continues *out of* Q beyond the corner of an unstable side, the image $F(Q_u)$ sticks out. Schematically (period two, as in Figure 10.27(a); solid: sides of the cell, bold red: unstable sides, dashed: F of the left unstable side):



Here $F(Q_u)$ runs past the top-left corner and leaves Q, so Q is a cell that is not a trapping region. A cell that is a trapping region is a *basin cell*; from now on Q is a basin cell and $B = \operatorname{bas}(Q)$.

Step 4: Accessibility is preserved by F and F^{-1}

Invariance of B and of $\operatorname{bd} B$. For any point x, $x \in B \iff F(x) \in B$: if some $x_n \in \operatorname{int} Q$ ($n \geq 0$) then $F(x)$ has $F(x)_{n-1} = x_n \in \operatorname{int} Q$ for $n \geq 1$ and, if $n = 0$, $x_1 = F(x_0) \in \operatorname{int} Q$ by

Step 1; conversely if $F(x)_m \in \operatorname{int} Q$ then $x_{m+1} \in \operatorname{int} Q$. Hence $F^{-1}(B) = B$, and since F is onto, $F(B) = B$. Because F is a homeomorphism,

$$F(\operatorname{bd} B) = \operatorname{bd} F(B) = \operatorname{bd} B, \qquad F^{-1}(\operatorname{bd} B) = \operatorname{bd} B.$$

So $\operatorname{bd} B$ is invariant under F (in both directions).

Accessibility. Let $x \in \operatorname{bd} B$ be accessible: there is a curve γ with $\gamma(0) = x$ and $\gamma((0,1]) \subset B$. Then $F \circ \gamma$ is a curve with endpoint $F(x) \in \operatorname{bd} B$ and $(F \circ \gamma)((0,1]) \subset F(B) = B$: $F(x)$ is accessible. Conversely, if $F(x)$ is accessible via γ, then $F^{-1} \circ \gamma$ is a curve with endpoint x and the rest in $F^{-1}(B) = B$, so x is accessible. Thus x is accessible $\iff$ $F(x)$ is accessible, and by induction

$$x \text{ accessible} \iff x_n \text{ accessible for every } n = 1, 2, 3, \ldots \qquad \blacksquare$$

Step 5: The set U and curves that avoid it

Proposition 1. *If $x \in \operatorname{bd} B$ is accessible from B, then $x \in W^s(\mathbf{p})$.*

Fix such an x. If x_n is a point of $\mathbf{p}$ for some n, then $x \in W^s(\mathbf{p})$ by (M1) and we are done; so assume no x_n lies in $\mathbf{p}$.

The set U. For each i let $U_{0,i}$ be the closed arc of the branch of $W^u(p_i)$ that runs from p_i along the loop to the far end e_i of the unstable side of p_i (so $U_{0,i}$ contains the closure of that unstable side and the connecting piece from p_i to it), and let $U_0 = \bigcup_i U_{0,i}$. This is the text's U. Let

$$U = \bigcup_{j=0}^{K-1} F^{-j}(U_0).$$

(This only adds finitely many preimage arcs.) Its properties:

(U1) U is compact (finite union of compact arcs), $U \subset W^u(\mathbf{p})$, and $U \supset Q_u$.

(U2) $F^{-1}(U) \subset U$. Indeed $F^{-1}(U) = \bigcup_{j=1}^{K} F^{-j}(U_0)$ and $F^{-K}(U_0) \subset U_0$ by (1) applied to each arc $U_{0,i}$.

(U3) For each a, $U \cap W^u(p_a)$ lies in two initial arcs (one on each branch) of $W^u(p_a)$, i.e. in a bounded range of the branch parameter.

Some iterate of x is outside U. Suppose $x_n \in U$ for all $n \geq 0$. Then $x \in U \subset W^u(p_a)$ for some a, with $x \neq p_a$, say at parameter $s > 0$ of a branch. By (1), $F^{Km}(x)$ has parameter $\to \infty$, so by (U3) $F^{Km}(x) \notin U$ for large m, a contradiction. Choose n_0 with $x_{n_0} \notin U$. By Step 4, $x_{n_0} \in \operatorname{bd} B$ is accessible. Replacing x by x_{n_0} (if the result holds for x_{n_0} then $x \in W^s(\mathbf{p})$ by (M1)), *we may assume $x \notin U$.*

Step 5 claim. *There is a curve J with endpoint x, $J \setminus \{x\} \subset B$, $J \cap U = \emptyset$, and $F^n(J) \cap U = \emptyset$ for all $n = 0, 1, 2, \ldots$*

Proof. Let γ be a curve with $\gamma(0) = x$ and $\gamma((0,1]) \subset B$. U is closed and $x \notin U$, so $d = \operatorname{dist}(x, U) > 0$; by continuity there is $t_0 > 0$ with $|\gamma(t) - x| < d$ for $t \leq t_0$. Put $J = \gamma([0, t_0])$: it has endpoint x, lies in B except for x, and misses U.

Now suppose $F^n(y) \in U$ for some $y \in J$ and $n \geq 0$. Then $y \in F^{-n}(U) \subset U$ by (U2), contradicting $J \cap U = \emptyset$. Hence $F^n(J) \cap U = \emptyset$. $\blacksquare$

Also, by Step 4, $F^n(J) \setminus \{x_n\} \subset F^n(B) = B$ and $x_n \in \operatorname{bd} B$ for every n.

Step 6: The curve is carried into Q; conclusion of Proposition 1

Take $y \in J$, $y \neq x$. Then $y \in B$, so by the definition of B there is n with $F^n(y) \in \operatorname{int} Q$. (This is why n can be chosen.)

Claim. $F^n(J) \subset Q$.

Proof. $F^n(J \setminus \{x\})$ is connected (continuous image of $(0,1]$) and contains $F^n(y) \in \operatorname{int} Q$. It avoids

- Q_u, since $F^n(J) \cap U = \emptyset$ and $Q_u \subset U$ (Step 5);
- Q_s, since $F^n(J\setminus\{x\}) \subset B$ while $Q_s \cap B = \emptyset$: indeed if $z \in Q_s$ then by (C3) $F^m(z) \in Q_s \subset \operatorname{bd} Q$ for all m, and $\operatorname{bd} Q \cap \operatorname{int} Q = \emptyset$, so no iterate of z enters $\operatorname{int} Q$.

So $F^n(J \setminus \{x\})$ is a connected set disjoint from $\operatorname{bd} Q = Q_s \cup Q_u$. The sets $\operatorname{int} Q$ and $\mathbb{R}^2 \setminus Q$ are disjoint open sets whose union is $\mathbb{R}^2 \setminus \operatorname{bd} Q$, and a connected set missing $\operatorname{bd} Q$ lies in one of them; it meets $\operatorname{int} Q$, hence $F^n(J \setminus \{x\}) \subset \operatorname{int} Q$. Finally $F^n(x) \in \operatorname{cl}(\operatorname{int} Q) \subset Q$. ■

$F^n(x) \in W^s(\mathbf{p})$. The point $x_n = F^n(x)$ satisfies:

- $x_n \in \operatorname{bd} B$ (invariance of $\operatorname{bd} B$);
- $x_n \in Q$ (the claim);
- $x_n \notin \operatorname{int} Q$, because $\operatorname{int} Q \subset B$ and B is open, so $B \cap \operatorname{bd} B = \emptyset$;
- $x_n \notin Q_u$, because $x_n \in F^n(J)$ and $F^n(J) \cap U = \emptyset$.

Hence $x_n \in Q \setminus \operatorname{int} Q = \operatorname{bd} Q$ and $x_n \notin Q_u$, so $x_n \in Q_s \subset W^s(\mathbf{p})$. By (M1), $x \in W^s(\mathbf{p})$. This proves Proposition 1. ■

By-product (used in Step 8). $Q_s \subset \operatorname{bd} B$ and, more generally, $W^s(\mathbf{p}) \subset \operatorname{bd} B$. *Proof.* By the argument above $Q_s \cap B = \emptyset$. Also $Q_s \subset \operatorname{bd} Q \subset \operatorname{cl}(\operatorname{int} Q) \subset \operatorname{cl} B$ (the boundary of a closed topological disk is in the closure of its interior). So $Q_s \subset \operatorname{cl} B \setminus B = \operatorname{bd} B$. If $y \in W^s(p_i)$, then $F^{Km}(y)$ eventually lies in the stable side σ_i (which contains a neighborhood of p_i in $W^s(p_i)$) and so in $Q_s \subset \operatorname{bd} B$; as $F^{-1}(\operatorname{bd} B) = \operatorname{bd} B$, $y \in \operatorname{bd} B$. ■

Step 7: Accessible points are dense in $\operatorname{bd} B$ (Proposition 2)

Let $z \in \operatorname{bd} B$ and $\varepsilon > 0$. We find an accessible point within ε of z.

Since $z \in \operatorname{bd} B$, the neighborhood $N_\varepsilon(z)$ contains a point $w \in B$. Consider the straight segment $\ell(\tau) = w + \tau(z - w)$, $\tau \in [0,1]$, and let

$$\tau^* = \sup\{\tau \in [0,1] :\ \ell(\sigma) \in B \text{ for all } \sigma \in [0,\tau]\}.$$

Since $w \in B$ and B is open, $\tau^* > 0$, and $\ell(\sigma) \in B$ for all $\sigma < \tau^*$. Let $a = \ell(\tau^*)$.

- $a \notin B$: if $\tau^* < 1$ and $a \in B$, openness of B would let us go a little further, contradicting the definition of τ^*; if $\tau^* = 1$ then $a = z \notin B$ since $z \in \operatorname{bd} B$ and B is open.
- $a \in \operatorname{cl} B$, being a limit of the points $\ell(\sigma) \in B$, $\sigma \uparrow \tau^*$.

Hence $a \in \operatorname{cl} B \setminus B = \operatorname{bd} B$. The curve $\tau \mapsto \ell(\tau^* - \tau\tau^*)$, i.e. the segment from a back to w, has endpoint a and all its other points in B: a is accessible. Finally a lies on the segment from w to z, so $|a - z| \leq |w - z| < \varepsilon$. Thus accessible points are dense in $\operatorname{bd} B$. ■

Step 8: Proposition 3 and Corollary 4

Let B_2 be another basin: $B_2 = \operatorname{bas}(Q_2)$ for a trapping region Q_2, so B_2 is open (Step 2) and $F^{-1}(B_2) = B_2$ (Step 4), and $B \cap B_2 = \emptyset$.

Proposition 3. *If $W^u(\mathbf{p})$ enters B_2 (i.e. $W^u(\mathbf{p}) \cap B_2 \neq \emptyset$), then $\operatorname{bd} B \subset \operatorname{bd} B_2$.*

Proof. **Part 1:** $W^s(\mathbf{p}) \subset \operatorname{bd} B_2$**.** Pick $q \in W^u(p_a) \cap B_2$. As B_2 is open and $W^u(p_a)$ is a smooth curve through q, there is a short arc $D \subset B_2$ through q transverse to $W^u(p_a)$. Apply the Lambda Lemma to $g = F^{-K}$, which has a saddle fixed point at p_a with $W^s_g(p_a) = W^u(p_a)$ and $W^u_g(p_a) = W^s(p_a)$: D crosses $W^s_g(p_a)$ transversally at q. So for every $y \in W^s(p_a)$ and $\delta > 0$ some $g^n(D) = F^{-Kn}(D)$ meets $N_\delta(y)$. Since $F^{-1}(B_2) = B_2$, $F^{-Kn}(D) \subset B_2$. Hence $y \in \operatorname{cl} B_2$. Applying this at each p_a in the orbit (the orbit of q under F moves q into $W^u(p_{a'}) \cap B_2$ for every a', since $F(W^u(p_a)) = W^u(p_{a+1})$ and $F(B_2) = B_2$) gives $W^s(\mathbf{p}) \subset \operatorname{cl} B_2$. By the by-product of Step 6, $W^s(\mathbf{p}) \subset \operatorname{bd} B$, and B_2 is an open set disjoint from B, hence disjoint from $\operatorname{cl} B \supset \operatorname{bd} B$. So $W^s(\mathbf{p}) \cap B_2 = \emptyset$ and

$$W^s(\mathbf{p}) \subset \operatorname{cl} B_2 \setminus B_2 = \operatorname{bd} B_2.$$

Part 2: all of $\operatorname{bd} B$**.** By Proposition 1, every accessible point of $\operatorname{bd} B$ lies in $W^s(\mathbf{p}) \subset \operatorname{cl} B_2$. By Proposition 2 these points are dense in $\operatorname{bd} B$, and $\operatorname{cl} B_2$ is closed, so $\operatorname{bd} B \subset \operatorname{cl} B_2$. Finally $\operatorname{bd} B \cap B_2 = \emptyset$ (B_2 open and disjoint from B), hence $\operatorname{bd} B \subset \operatorname{cl} B_2 \setminus B_2 = \operatorname{bd} B_2$. ■

Corollary 4. *If $W^u(\mathbf{p})$ enters two other basins B_2 and B_3, with B, B_2, B_3 pairwise disjoint, then every boundary point of B is also a boundary point of B_2 and of B_3; so B is a Wada basin.*

Proof. Apply Proposition 3 once with B_2 and once with B_3: $\operatorname{bd} B \subset \operatorname{bd} B_2$ and $\operatorname{bd} B \subset \operatorname{bd} B_3$. So each point of $\operatorname{bd} B$ is in the boundary of all three basins, which is exactly the definition of a Wada basin. ■

Remark. (1) In the pendulum picture of Chapter 2 each of the three basins has a basin cell whose generating orbit has unstable manifold entering all three basins; Corollary 4 applied to each basin shows $\operatorname{bd} B_1 = \operatorname{bd} B_2 = \operatorname{bd} B_3$: the three basins have a *common boundary*, each boundary point being a "shore point" of all three lakes, as in the Lakes of Wada. (2) Besides the Stable Manifold and Lambda Lemma facts quoted, the proofs used: the fact $F(\operatorname{bd} Q) \subset Q \Rightarrow F(Q) \subset Q$ (given in the text), the property (C3) $F(Q_s) \subset Q_s$ of the stable sides of a cell, and that the basin B_2 is open, F-invariant and disjoint from B.

Summary

Step	Result
1	$F(Q) \subset Q \Rightarrow x_j \in Q$ implies $x_n \in Q$ for $n \geq j$ (also with $\operatorname{int} Q$).
2	$\operatorname{bas}(Q) = \bigcup_n F^{-n}(\operatorname{int} Q)$ is open; path connected when $\operatorname{int} Q$ is (the literal hypothesis "Q path connected" is not enough: two tangent disks, $F = x/2$).
3	A cell is trapping iff $F(Q_u) \subset Q$ ($F(Q_s) \subset Q_s$ is automatic); sketch of a non-trapping cell.
4	$\operatorname{bd} B$ is F-invariant; x accessible $\iff$ x_n accessible.
5	Replace J by a short tail avoiding the closed, backward-invariant set U; then $F^n(J) \cap U = \emptyset$.
6	$F^n(J) \subset Q$, $x_n \in Q_s$, so $x \in W^s(\mathbf{p})$ (Proposition 1).
7	Accessible points are dense in $\operatorname{bd} B$ (stop at the first boundary point on a segment from B).
8	Lambda Lemma: $W^u(\mathbf{p})$ meeting B_2 forces $W^s(\mathbf{p}) \subset \operatorname{bd} B_2$, then $\operatorname{bd} B \subset \operatorname{bd} B_2$; two such basins give a Wada basin.